Prayas JEE AIR (2025)

WEEKLY TEST 28

DURATION : 180 Minutes | **DATE : 09/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

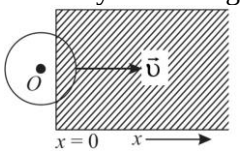
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Candidate's Signature: _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

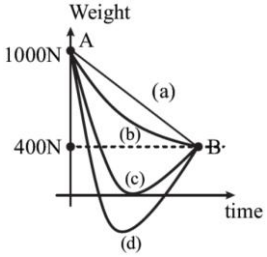
Single Correct Type Questions

1. A constant magnetic field of 1T is applied in the $x > 0$ region. A metallic circular ring of radius 1m is moving with a constant velocity of 1 m/s along the x-axis. At $t = 0$, the centre of O of the ring is at $x = -1$m. What will be the value of the induced emf in the ring at $t = 1$ s?
(Assume the velocity of the ring does not change.)



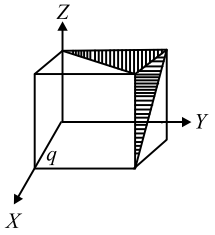
(1) 0V (2) 2πV
(3) 2V (4) 1V

2. A person whose mass is 100 kg travels from Earth to Mars in a spaceship. Neglect all other object in sky and take acceleration due to gravity on the surface of the Earth and mars as 10 m/s^2 and 4 m/s^2 respectively. Identify from the below figures, the curve that fits best for the weight of the passenger as a function of time.



(1) (c) (2) (a)
(3) (d) (4) (b)

3. A charge 'q' is placed at one corner of a cube as shown in figure. The flux of electrostatic field E though the shaded area is:



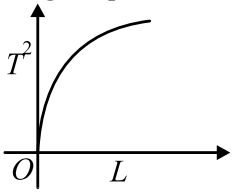
(1) $\frac{q}{48\in_0}$ (2) $\frac{q}{8\in_0}$
(3) $\frac{q}{24\in_0}$ (4) $\frac{q}{4\in_0}$

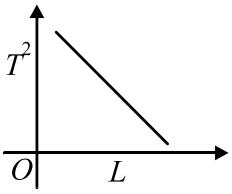
4. Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density d. The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other x_2. When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is:

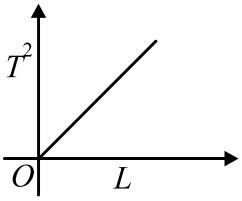
(1) $\frac{3}{4}gdS(x_2 - x_1)^2$ (2) $gdS(x_2 + x_1)^2$
(3) $\frac{1}{4}gdS(x_2 - x_1)^2$ (4) $gdS\left(x_2^2 + x_1^2\right)$

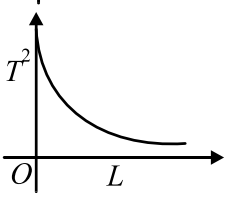
5. Given below are two statements: One is labelled as Assertion **A** and the other is labelled as Reason **R**.
Assertion (A): Two metallic spheres are charged to the same potential. One of them is hollow and another is solid, and both have the same radii. Solid sphere will have lower charge than the hollow one.
Reason (R): Capacitance of metallic spheres depend on the radii of spheres.
In the light of the above statements, choose the correct answer from the options given below.
(1) **A** is false but **R** is true
(2) Both **A** and **R** are true and **R** is the correct explanation of **A**
(3) A is true but **R** is false
(4) Both A and **R** are true but **R** is not the correct explanation of **A**

6. Choose the correct length (L) versus square of time period (T^2) graph for a simple pendulum executing simple harmonic motion.

(1) 

(2) 

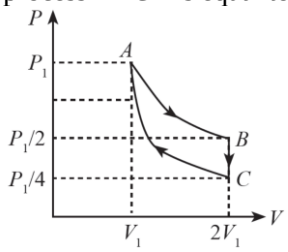
(3) 

(4) 

7. The length of metallic wire is l_1 when tension in it is T_1. It is l_2 when the tension is T_2. The original length of the wire will be:

(1) $\frac{T_2l_1 + T_1l_2}{T_1 + T_2}$ (2) $\frac{T_1l_1 - T_2l_2}{T_2 + T_1}$
(3) $\frac{T_2l_1 - T_1l_2}{T_2 - T_1}$ (4) $\frac{T_1l_1 + l_2T_2}{T_2 - T_1}$

8. If one mole of an ideal gas at (P_1, V_1, T) is allowed to expand reversibly and isothermally (A to B) its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value ($B \rightarrow C$). Then it is restored to its initial state by a reversible adiabatic compression (C to A). The net work done by the gas during the process ABCA is equal to:



(1) 0

(2) $RT \ln 2$

(3) $RT\left(\ln 2 - \frac{1}{2(\gamma-1)}\right)$

(4) $-\frac{RT}{2(\gamma-1)}$

9. There is a small source of light at some depth below the surface of water (refractive index = 4/3) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly): [Use the fact that surface area of spherical cap of height h and radius of curvature r is $2\pi rh$]

(1) 21% (2) 17%

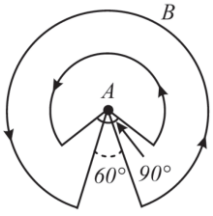
(3) 50% (4) 34%

10. The linear mass density of a thin rod AB of length L varies from A to B as $\lambda(x) = \lambda_0\left(1+\frac{x}{L}\right)$, where x is the distance from A. If M is the mass of the rod then its moment of inertia about an axis passing through A and perpendicular to the rod is:

(1) $\frac{5}{12}ML^2$ (2) $\frac{7}{18}ML^2$

(3) $\frac{2}{5}ML^2$ (4) $\frac{3}{7}ML^2$

11. A wire A, bent in the shape of an arc of a circle, carrying a current of 2A and having radius 2 cm and another wire B, also bent in the shape of arc of a circle, carrying a current of 3A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common centre O is

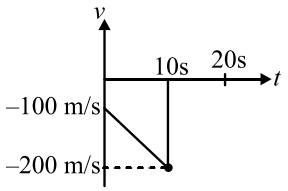


(1) 2 : 5 (2) 6 : 5

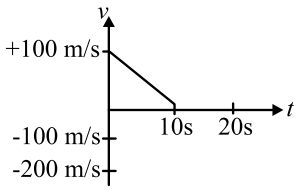
(3) 4 : 6 (4) 6 : 4

12. A bullet is shot vertically downwards with an initial velocity of 100 m/s from a certain height. Within 10s, the bullet reaches the ground and instantaneously comes to rest due to the perfectly inelastic collision. The velocity -time curve for total time t = 20 s will be: (Take g = 10m/s^2).

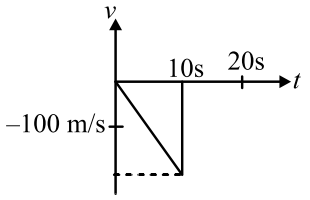
(1)



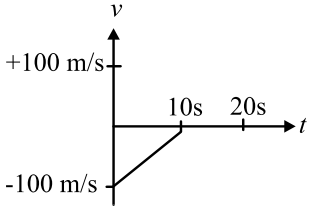
(2)



(3)



(4)



13. A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is true for measuring value of G?

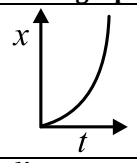
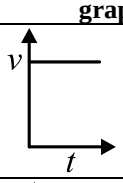
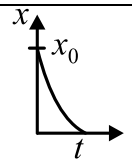
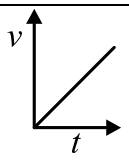
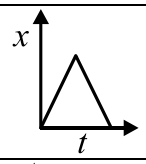
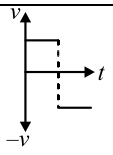
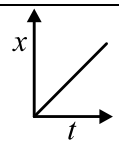
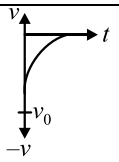
(1) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.

(2) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).

(3) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance(s).

(4) $\frac{1}{3}$ deflection method can be used and in this case, the G value equals to the shunt resistance(s).

14. Match Column-I with Column-II:

	Column-I (x-t graphs) (Position-Time graphs)		**Column-II (v-t graphs) (Velocity-Time graphs)**
A		P	
B		Q	
C		R	
D		S	

Choose the correct answer from the options given below:

(1) A→ Q; B→S; C→R; D→P
(2) A→ P; B→Q; C→R; D→S
(3) A→ Q; B→R; C→S; D→P
(4) A→ P; B→R; C→S; D→Q

15. A particle '*P*' is formed due to a completely inelastic collision of particles '*x*' and '*y*' having de-Broglie wavelengths 'λ_x' and 'λ_y' respectively. If *x* and *y* were moving in opposite directions, then the de-Broglie wavelength of '*P*' is

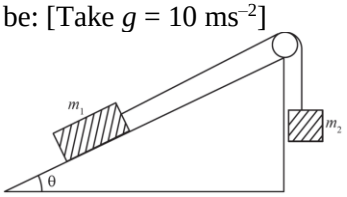
(1) $\lambda_x + \lambda_y$ (2) $\frac{\lambda_x\lambda_y}{\lambda_x + \lambda_y}$

(3) $\frac{\lambda_x\lambda_y}{\lambda_x - \lambda_y}$ (4) $\lambda_x - \lambda_y$

16. Imagine that the electron in a hydrogen atom is replaced by a muon (μ). The mass of muon particle is 207 times that of an electron and charge is equal to the charge of an electron. The ionization potential of this hydrogen atom will be

(1) 13.6 eV (2) 2815.2 eV
(3) 331.2 eV (4) 27.2 eV

17. Two bodies of masses m_1 = 5kg and m_2 = 3kg are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be: [Take g = 10 ms^{-2}]



(1) 30 N (2) 40 N
(3) 50 N (4) 60 N

18. Using Young's double slit experiment, a monochromatic light of wavelength 5000 Å produces fringes of fringe width 0.5 mm. If another monochromatic light of wavelength 6000 Å is used and the separation between the slits is doubled, then the new fringe width will be

(1) 0.5 mm (2) 1.0 mm
(3) 0.6 mm (4) 0.3 mm

19. A massless spring (k = 800 N/m), attached with a mass (500 g) is completely immersed in 1 kg of water. The spring is stretched by 2 cm and released so that it starts vibrating. What would be the order of magnitude of the change in the temperature of water when the vibrations stop completely? (Assume that the water container and spring receive negligible heat and specific heat of mass = 400 J/kg K, specific heat of water = 4184 J/kg K)

(1) 10^{-3} K (2) 10^{-4} K
(3) 10^{-1} K (4) 10^{-5} K

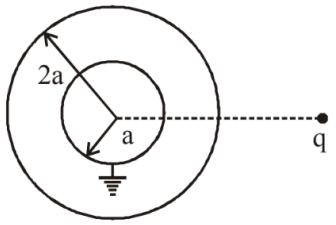
20. Given the masses of various atomic particles m_p = 1.0072 *u*, m_n = 1.0087 *u*, m_e = 0.000548 *u*, m_v = 0, m_d = 2.0141 *u*, where *p*→proton, *n*→neutron, *e* → electron, $\bar{\nu}$ → antineutrino and *d*→deuteron. Which of the following process is allowed by momentum and energy conservation?

(1) $n + n \rightarrow$ deuterium atom (electron bound to the nucleus)
(2) $n + p \rightarrow d + \gamma$
(3) $p \rightarrow n + e^+ + \bar{\nu}$
(4) $e^+ + e^- \rightarrow \gamma$

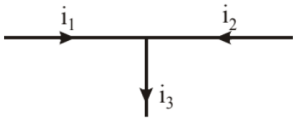
Integer Type Questions

21. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. Then, the lowest resonant frequency (Hz) for this string is

22. A solid conducting sphere of radius *a* is surrounded by a thin uncharged concentric conducting shell of radius 2*a*. A point charge *q* is placed at a distance 4*a* from common centre of conducting sphere and shell. The inner sphere is then grounded. The charge on solid sphere is $\frac{q}{x}$. Find the value of *x*.



23. If $i_1 = 3 \sin \omega t$, $i_2 = 4 \cos \omega t$, and $i_3 = i_0 \sin(\omega t + 53°)$, find the value of i_0.



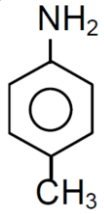
24. A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is: $x \times 10^{15}$. Find x.
[Given planck's constant $h = 6.6 \times 10^{-34}$ Js, speed of light $c = 3.0 \times 10^8$ m/s]

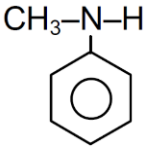
25. Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K respectively. The ratio of the energy radiated per second by the first sphere to that by the second is $x:y$. Find $[x + y]$ where x and y are nearest integer

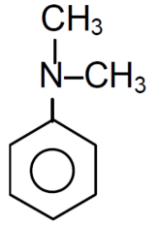
SECTION-II (CHEMISTRY)

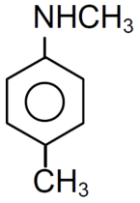
Single Correct Type Questions

26. Which of the following amine will show carbylamine reaction?

(1) 

(2) 

(3) 

(4) 

27. A hydrocarbon 'X' with formula C_6H_8 uses two moles of H_2 on catalytic hydrogenation of its one mole. On ozonolysis, 'X' yields two moles of methane dicarbaldehyde. The hydrocarbon 'X' is :

(1) hexa-1, 3, 5-triene
(2) 1-methylcyclopenta-1, 4-diene
(3) cyclohexa-1, 3-diene
(4) cyclohexa-1, 4-diene

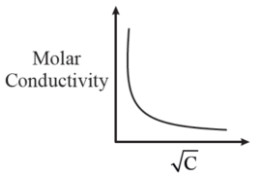
28. On increasing temperature ionic product of water increases [$H_2O \rightleftharpoons H^+ + OH^-$, $K_W = 10^{-14}$ at 25°C] which of the following statement is correct regarding dissociation of water.

(1) water becomes basic on decreasing the temperature
(2) pH of water increase on increasing the temperature
(3) water will be neutral irrespective of temperature change
(4) pH cannot be predicted on given information

29. Which of the following statements is correct?

(1) Gluconic acid is a dicarboxylic acid
(2) Gluconic acid can form cyclic (acetal/hemiacetal) structure
(3) Gluconic acid is a partial oxidation product of glucose
(4) Gluconic acid is obtained by oxidation of glucose with HNO_3

30. The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure



The electrolyte X is

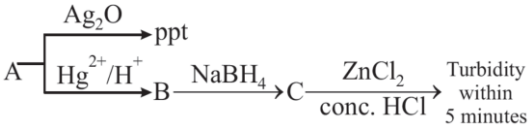
(1) NaCl (2) HCl
(3) CH_3COOH (4) KNO_3

31. Arrange the following solutions in the decreasing order of pOH

(a) 0.01 M HCI
(b) 0.01 M NaOH
(c) 0.01 M CH_3COONa
(d) 0.01 M NaCI

(1) (a) > (c) > (d) > (b)
(2) (a) > (d) > (c) > (b)
(3) (b) > (c) > (d) > (a)
(4) (b) > (d) > (c) > (a)

32. Consider the following reactions:



'A' is:

(1) CH ≡ CH (2) $CH_3 - C \equiv CH$
(3) $CH_2 = CH_2$ (4) $CH_3 - C \equiv C - CH_3$

33. The d–electron configuration of $[Ru(en)_3]Cl_2$ and $[Fe(H_2O)_6]Cl_2$, respectively are:

(1) $t_{2g}^6 e_g^0$ and $t_{2g}^4 e_g^2$ (2) $t_{2g}^4 e_g^2$ and $t_{2g}^4 e_g^2$

(3) $t_{2g}^6 e_g^0$ and $t_{2g}^6 e_g^0$ (4) $t_{2g}^4 e_g^2$ and $t_{2g}^6 e_g^0$

34. A compound 'X' when treated with phthalic anhydride in presence of concentrated H_2SO_4 yields 'Y'. 'Y' is used as an acid/base indicator. 'X' and 'Y' are respectively:

(1) Carbolic acid, Phenolphthalein
(2) Anisole, methyl orange
(3) Salicylaldehyde, Phenolphthalein
(4) Toludine, Phenolphthalein

35. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Metallic character decreases and non-metallic character increases on moving from left to right in a period.

Reason (R): It is due to increase in ionisation enthalpy and decrease in electron gain enthalpy, when one moves from left to right in a period.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) (A) is false but (R) is true
(2) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(4) (A) is true but (R) is false

36. Presence of which reagent will affect the reversibility of the following reaction, and change it to a irreversible reaction:

$$CH_4 + I_2 \underset{\text{Reversible}}{\overset{h\nu}{\rightleftharpoons}} CH_3I + HI$$

(1) HOCl (2) Liquid NH_3
(3) Dilute HNO_2 (4) Concentrated HIO_3

37. Elevation in the boiling point for 1 molal solution of glucose is 2 K. The depression in the freezing point for 2 molal solution of glucose in the same solvent is 2 K. The relation between K_b and K_f is

(1) $K_b = 0.5\ K_f$ (2) $K_b = 2K_f$
(3) $K_b = 1.5\ K_f$ (4) $K_b = K_f$

38. Outermost electronic configuration of a group 13 element, E, is $4s^2$, $4p^1$. The electronic configuration of an element of p-block period-five placed diagonally to element, E is:

(1) [Kr] $4d^{10}\,5s^2\,5p^2$ (2) [Kr] $3d^{10}\,4s^2\,4p^2$
(3) [Xe] $5d^{10}\,6s^2\,6p^2$ (4) [Ar] $3d^{10}\,4s^2\,4p^2$

39. The increasing order of the reactivity of the following compounds in nucleophilic addition reaction is:

Propanal, Benzaldehyde, Propanone, Butanone

(1) Benzaldehyde < Propanal< Propanone < Butanone
(2) Butanone < Propanone < Benzaldehyde < Propanal
(3) Propanal< Propanone < Butanone <Benzaldehyde
(4) Benzaldehyde < Butanone < Propanone < Propanal

40. Match Column-I with Column-II

	Column-I		**Column-II**
A	XeF_4	P	See-saw
B	SF_4	Q	Square planar
C	NH_4^+	R	Bent T-shaped
D	BrF_3	S	Tetrahedral

Choose the correct answer from the options given below:

(1) A→S; B→R; C→Q; D→P
(2) A→Q; B→P; C→R; D→S
(3) A→S; B→P; C→Q; D→R
(4) A→Q; B→P; C→S; D→R

41. The correct statement is:

(1) Rate constant increases if activation energy increases
(2) If a reaction has zero activation energy, its rate is independent of temperature.
(3) The stronger the temperature dependence of the rate constant, the smaller is the activation energy.
(4) If there is no correlation between the temperature and the rate constant then it means that the reaction has negative activation energy.

42. Which of the following is NOT a correct method of the preparation of benzylamine from cyanobenzene?

(1) (i) HCl/H_2O (ii) $NaBH_4$
(2) (i) $LiAlH_4$ (ii) H_3O^+
(3) (i) $SnCl_2$+HCl(gas) (ii) $NaBH_4$
(4) H_2/Ni

43. The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is

(1) – COOH > – $CONH_2$ > – COCl > – CHO
(2) – SO_3H > – COCl > – $CONH_2$ > – CN
(3) – COOR > – COCl > – NH_2 > C = O
(4) – COOH > – COOR > – $CONH_2$ > – COCl

44. The incorrect expression among the following is:

(1) $\frac{\Delta G_{system}}{\Delta S_{total}} = -T$

(2) In an isothermal process,

$$W_{reversible} = -nRT\ln\frac{V_2}{V_1}$$

(3) $\ln K = \frac{\Delta H_0 - T\Delta S^\circ}{RT}$

(4) $K = e^{-\Delta G^\circ/RT}$

45. The correct order of following 3d metal oxides, according to their oxidation numbers is:

(A) CrO_3

(B) Fe_2O_3

(C) MnO_2

(D) V_2O_5

(E) Cu_2O

(1) (A) > (D) > (C) > (B) > (E)

(2) (C) > (A) > (D) > (E) > (B)

(3) (D) > (A) > (B) > (C) > (E)

(4) (A) > (C) > (D) > (B) > (E)

Integer Type Questions

46. In alkaline medium, how many moles of potassium permanganate are required for complete oxidation of 3.5 moles of KI to KIO_3?

47. Zinc reacts with hydrochloric acid to give hydrogen and zinc chloride. The volume of hydrogen gas produced at STP from the reaction of 11.5 g of zinc with excess HCl is _____ L (Nearest integer)
(Given: Molar mass of Zn is 65.4g mol^{-1} and Molar volume of H_2 at STP = 22.4L)

48. Dissociation constant for a weak base BOH is 10^{-5}. What is the value of pH when it is half neutralized by HCl?

49. If ΔG° for the half cell MnO_4^- / MnO_2 in an acid solution is -xF, then find the value of x

$$\left(\text{Given}: E^\circ_{MnO_4^-/Mn^{2+}} = 1.5\,V;\ E^\circ_{MnO_2/Mn^{2+}} = 1.25\,V\right)$$

50. For the reaction A (l) → 2B (g)
ΔU = 2.1 kcal, ΔS = 20 cal K^{-1} at 300 K
If the value of $|\Delta G|$ is x then find the value of 100x.

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which of the following statements is not true?

(1) $p_5 = p_2 \cdot p_3$

(2) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

(3) $p_3 = p_5 - p_4$

(4) $p_5 = 11$

52. Let $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$, where $a, c \in R$. if $A^3 = A$ and the positive value of a belongs to the interval $(n - 1, n)$ where $n \in N$, then n is equal to

(1) 4

(2) 2

(3) 3

(4) 1

53. The positive value of λ for which the co-efficient of x^2 in the expression $x^2\left(\sqrt{x} + \frac{\lambda}{x^2}\right)^{10}$ is 720, is:

(1) 4

(2) $2\sqrt{2}$

(3) $\sqrt{5}$

(4) 3

54. Let $A = \left\{z \in C : \left|\frac{z+1}{z-1}\right| < 1\right\}$ and

$$B = \left\{z \in C : \arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}\right\}$$

Then $A \cap B$ is

(1) A portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

(2) A portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

(3) An empty set

(4) A portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in the third quadrant only

55. Let the circumcentre of a triangle with vertices A (a, 3), B (b, 5) and C (a, b), $ab > 0$ be P (1, 1). If the line AP intersects the line BC at the point Q (k_1, k_2), then $k_1 + k_2$ is equal to:

(1) 2 (2) $\frac{4}{7}$

(3) $\frac{2}{7}$ (4) 4

56. Let $S_1 : x^2 + y^2 = 9$ and $S_2 : (x-2)^2 + y^2 = 1$. Then the locus of centre of a variable circle S which touches S_1 internally and S_2 externally always passes through the points.

(1) $\left(\frac{1}{2}, \pm\frac{\sqrt{5}}{2}\right)$ (2) $\left(0, \pm\sqrt{3}\right)$

(3) $(1, \pm 2)$ (4) $\left(2, \pm\frac{3}{2}\right)$

57. If $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}\,dx = g(x) + c,\ g(1) = 0,$ then $g\left(\frac{1}{2}\right)$ is equal to

(1) $\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right) + \frac{\pi}{3}$

(2) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) + \frac{\pi}{3}$

(3) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) - \frac{\pi}{3}$

(4) $\frac{1}{2}\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) - \frac{\pi}{6}$

58. The sum $\sum_{k=1}^{20}(1+2+3+....+k)$ is

(1) 1440 (2) 1340

(3) 1240 (4) 1540

59. The length of the latus rectum of a parabola, whose vertex and focus are on the positive x-axis at a distance R and $S(>R)$ respectively from the origin, is:

(1) $4(S+R)$ (2) $2(S+R)$

(3) $2(S-R)$ (4) $4(S-R)$

60. Let the functions $f : R \to R$ and $g : R \to R$ be defined as:

$$f(x) = \begin{cases} x+2, & x<0 \\ x^2, & x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} x^3, & x<1 \\ 3x-2, & x \geq 1 \end{cases}$$

Then, the number of points in R where $(fog)(x)$ is NOT differentiable is equal to:

(1) 0 (2) 1

(3) 2 (4) 3

61. The number of integers, greater than 7000 that can be formed, using the digits 3, 5,6,7, 8 without repetition, is

(1) 120 (2) 168

(3) 220 (4) 48

62. A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is:

(1) $\frac{1}{\sqrt{7}}$ (2) $2\sqrt{2}-1$

(3) $\frac{1}{2\sqrt{2}}$ (4) $\sqrt{7}-1$

63. The number of elements in the set $\{n \in Z: |n^2 - 10n + 19| < 6\}$

(1) 8 (2) 2

(3) 7 (4) 6

64. The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with eccentricity:

(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{2\sqrt{2}}$

(3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{2}$

65. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k},\ \vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. If a vector $\vec{d}$ satisfies $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{d}.\vec{a} = 24,$ then $|\vec{d}|^2$ is equal to

(1) 413 (2) 423

(3) 323 (4) 313

66. If the vertices of a hyperbola be at (-2, 0) and (2, 0) and one of its foci be at (-3,0), then which one of the following points does not lie on this hyperbola?

(1) $\left(-6, 2\sqrt{10}\right)$ (2) $\left(2\sqrt{6}, 5\right)$

(3) $\left(4, \sqrt{15}\right)$ (4) $\left(6, 5\sqrt{2}\right)$

67. Let the solution curve $y = y(x)$ of the differential equation

$$\left[\frac{x}{\sqrt{x^2-y^2}} + e^{\frac{y}{x}}\right]x\frac{dy}{dx} = x + \left[\frac{x}{\sqrt{x^2-y^2}} + e^{\frac{y}{x}}\right]y$$

pass through the points (1, 0) and $(2\alpha, \alpha)$, $\alpha > 0$. Then α is equal to

(1) $\frac{1}{2}\exp\left(\frac{\pi}{6} + \sqrt{e} - 1\right)$

(2) $\frac{1}{2}\exp\left(\frac{\pi}{3} + e - 1\right)$

(3) $\exp\left(\frac{\pi}{6} + \sqrt{e} + 1\right)$

(4) $2\exp\left(\frac{\pi}{3} + \sqrt{e} - 1\right)$

68. Let $f:[-3,1]\to R$ be given as

$$f(x)=\begin{cases}\min\{(x+6),x^2\}, & -3\le x\le 0\\ \max\left(\sqrt{x},x^2\right), & 0\le x\le 1\end{cases}$$

It the area bounded by $y=f(x)$ and x-axis is A, then the value of $6A$ is equal to ____.

(1) 21 (2) 81

(3) 41 (4) 51

69. If m and n respectively are the number of local maximum and local minimum points of the function $f(x)=\int_0^{x^2}\frac{t^2-5t+4}{2+e^t}dt$, then the ordered pair (m, n) is equal to

(1) (3, 2)

(2) (2, 3)

(3) (3, 5)

(4) (3, 4)

70. The value of $2\sin(12°)-\sin(72°)$ is:

(1) $\frac{\sqrt{5}\left(1-\sqrt{3}\right)}{4}$

(2) $\frac{1-\sqrt{5}}{8}$

(3) $\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{2}$

(4) $\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{4}$

Integer Type Questions

71. The mean of 10 numbers 7×8, 10×10, 13×12, 16×14, ….. is ____.

72. If the lines $\frac{x-1}{2}=\frac{2-y}{-3}=\frac{z-3}{\alpha}$ and $\frac{x-4}{5}=\frac{y-1}{2}=\frac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is

73. $f(x)=\int\frac{5x^8+7x^6}{(x^2+1+2x^7)^2}dx$, $(x\ge 0)$, $f(0)=0$ and $f(1)=\frac{1}{K}$ then the value of K is.

74. A student appeared in an examination consisting of 8 true - false type questions. The student guesses the answers with equal probability. The smallest value of n, so that the probability of guessing at least 'n' correct answers is less than $\frac{1}{2}$, is

75. If the coefficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11}$ and the coefficient of x^{-9} in $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to



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